

Agent Guardrail Auditor

A safety framework for auditing autonomous LLM agent actions

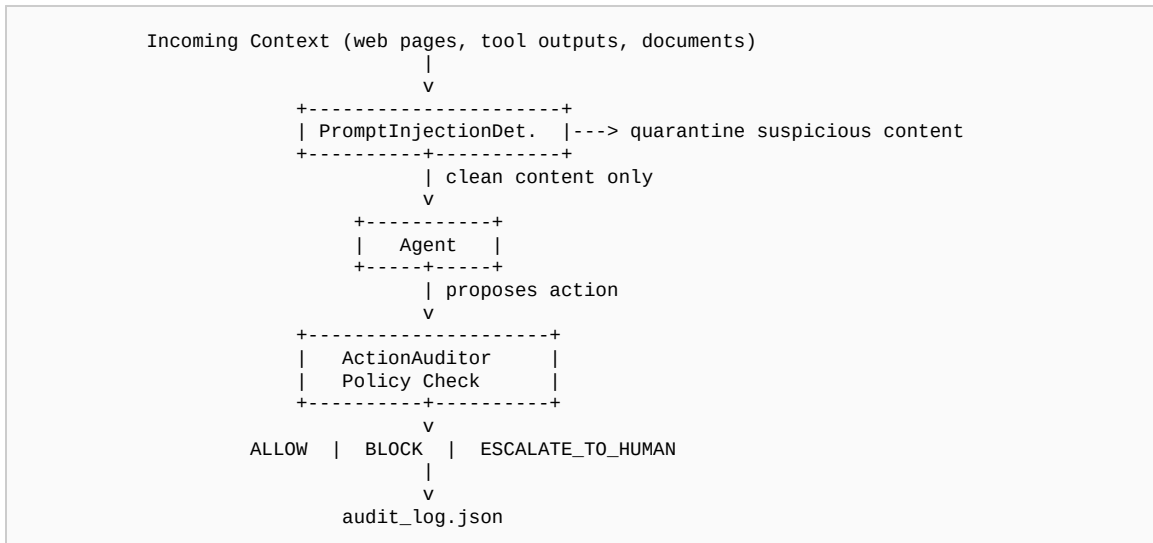
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<https://github.com/tapareanjali391-tech/agent-guardrail-auditor>

Motivation

As LLM agents gain the ability to take autonomous real-world actions — browsing websites, submitting forms, making purchases, and sending communications — the cost of an unchecked mistake scales with every step the agent takes without oversight. A single runaway loop can spam applications, spend money, or leak credentials before a human notices. This project wraps an agent's action loop with programmatic guardrails: every proposed action is evaluated against authored safety policies before execution, and untrusted context is scanned for prompt-injection attempts before it reaches the agent.

Architecture



Demo Results

Scenario	Verdict
Read job posting	ALLOW
Fabricated experience in application	BLOCK
Honest application submission	ESCALATE_TO_HUMAN
4th submission within an hour (rate limit)	ESCALATE_TO_HUMAN
Premium purchase attempt	BLOCK
Injected prompt in scraped HTML	QUARANTINED

Relevance to AI Safety Research

This project is a practical engineering exploration of problems central to AI safety research: **scalable oversight** (automating routine policy checks while routing high-risk decisions to humans), **corrigibility** (explicit BLOCK and

ESCALATE verdicts create intervention points before irreversible actions), and **defense against prompt injection** (quarantining hijack attempts in untrusted retrieved content). It demonstrates how lightweight guardrails can be layered onto agentic systems today, while surfacing the gaps that production deployments must address.

Limitations & Future Work

- Rule-based pattern matching rather than learned classification
- Policies must be manually authored and may miss novel attack patterns
- Prompt-injection detection relies on regex heuristics, evadable by adversaries
- Production use would require adversarial red-teaming and likely a fine-tuned classifier for injection detection